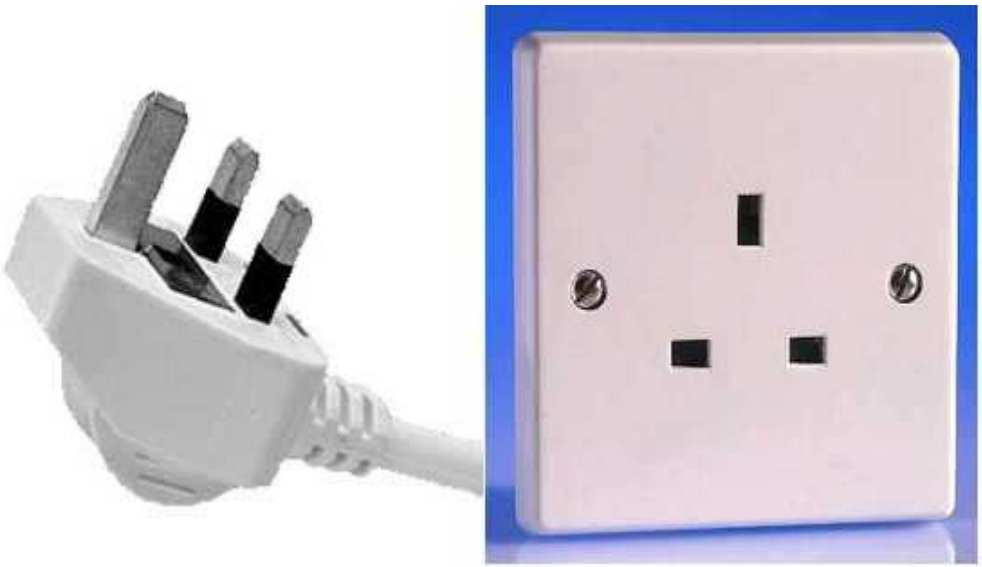
PHILIPS

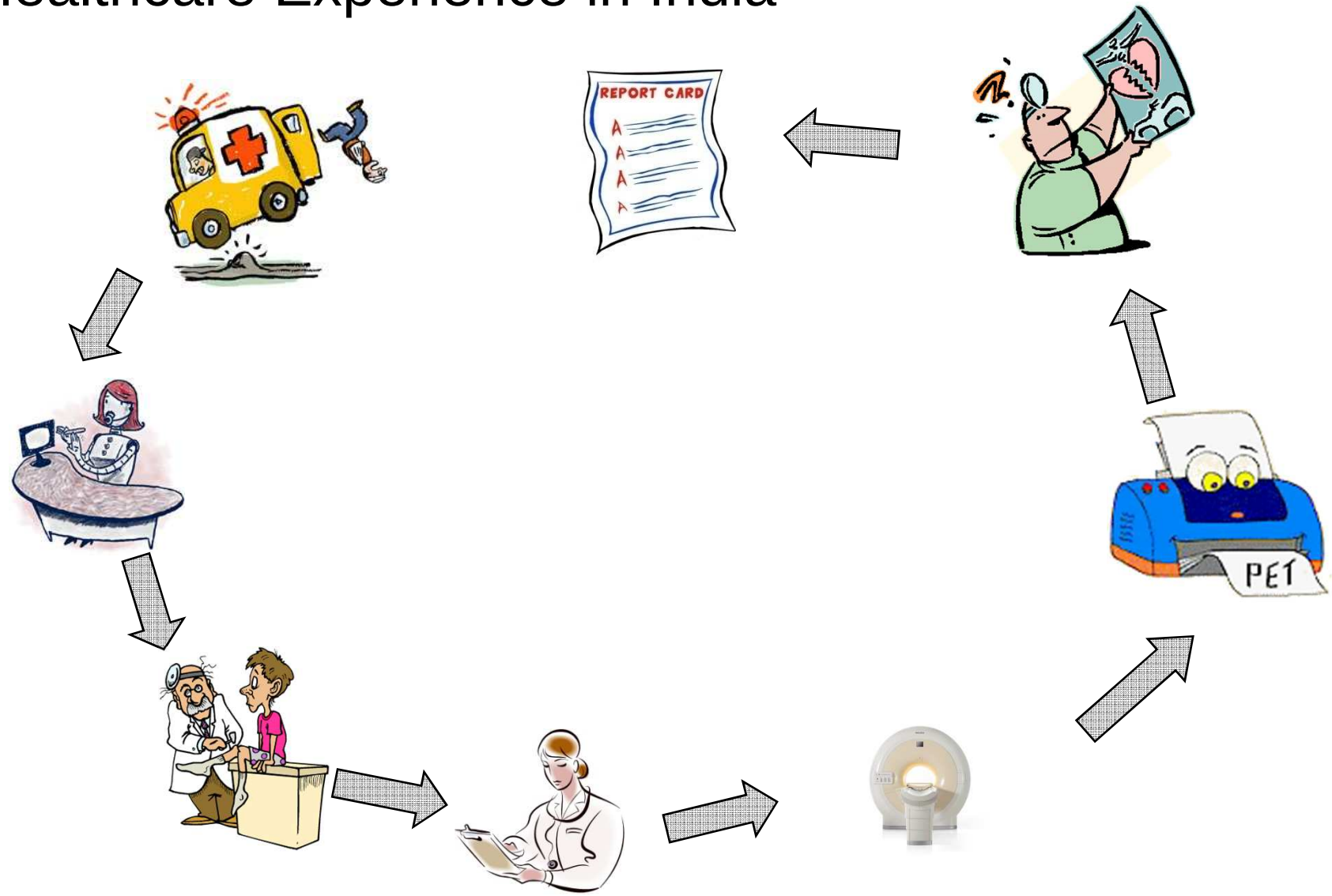
DICOM Introduction

Agenda

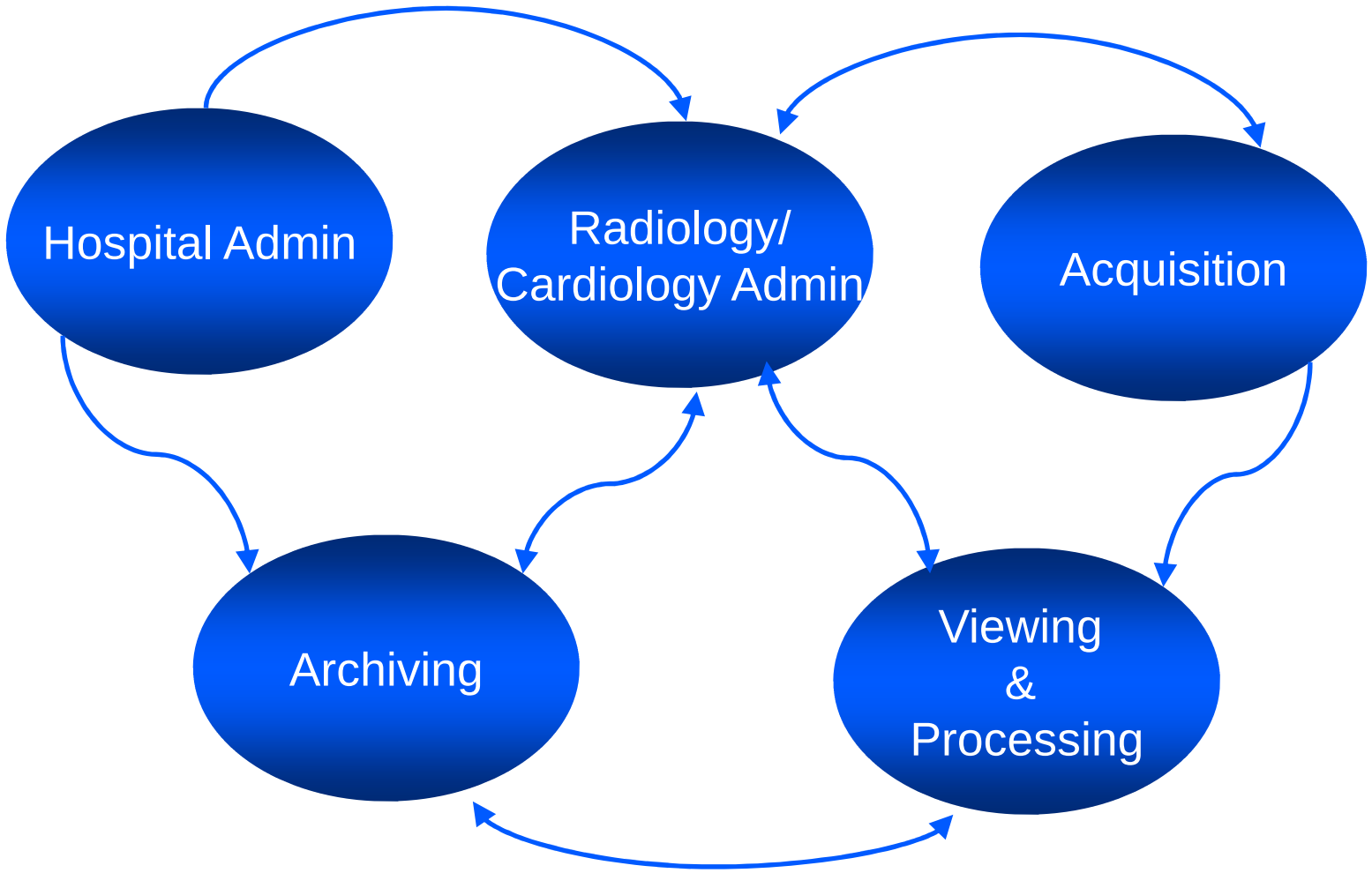
- Need for Interoperability
- Standards for Healthcare
- History of DICOM
- What is DICOM



Healthcare Experience in India



Domains within Healthcare

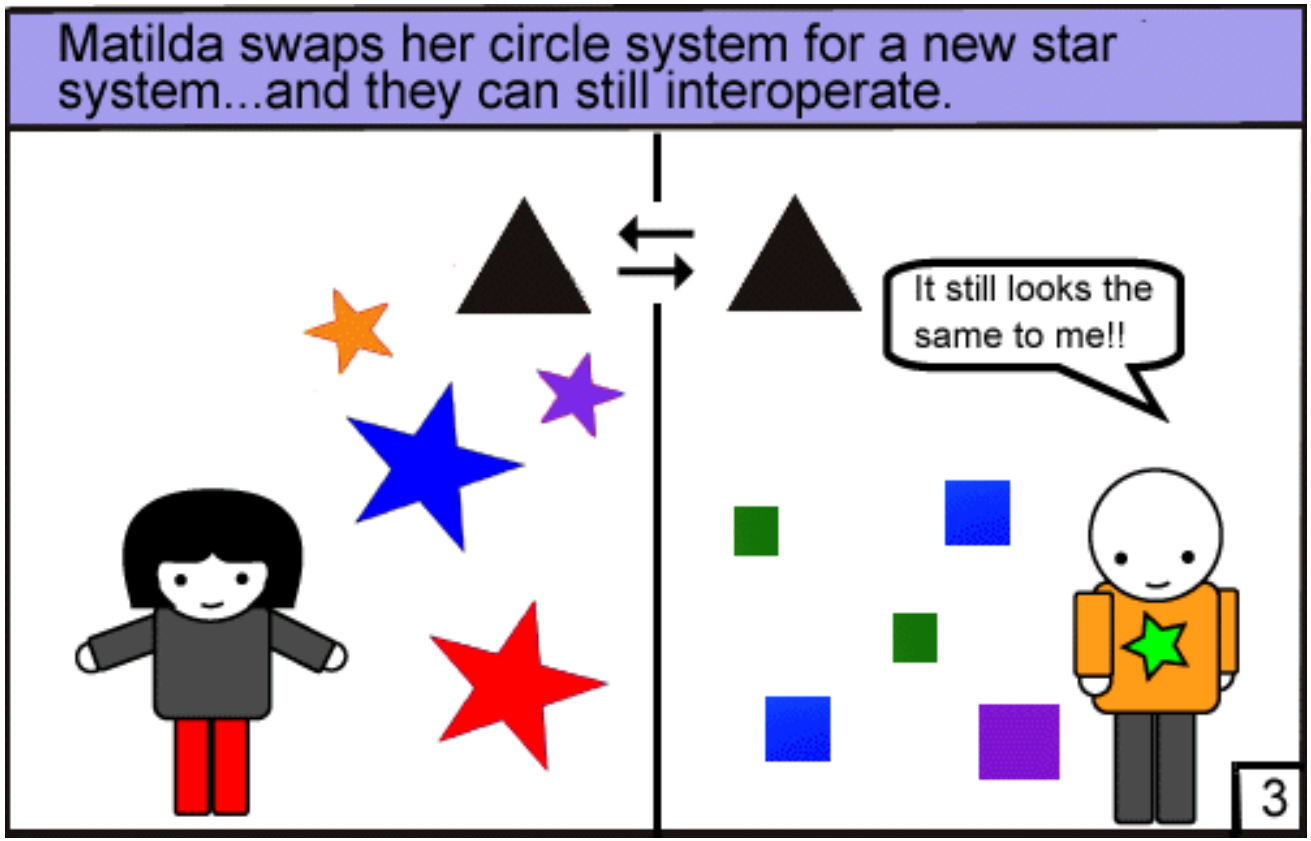


Challenges in Healthcare Environment

- Different Types of Medical Equipment (HIS, CIS, Modality, Hemodynamic systems, PACS ….)
- Manufactured by Different Vendors (Philips, GE, Siem
- Producing Different Types of Im
- Using
- Diagn ysicians (Technician, Cardiologist …)
- Loca ecific Laws & Regulations (US, Europe, Asia …)

Interoperability

Communication Story

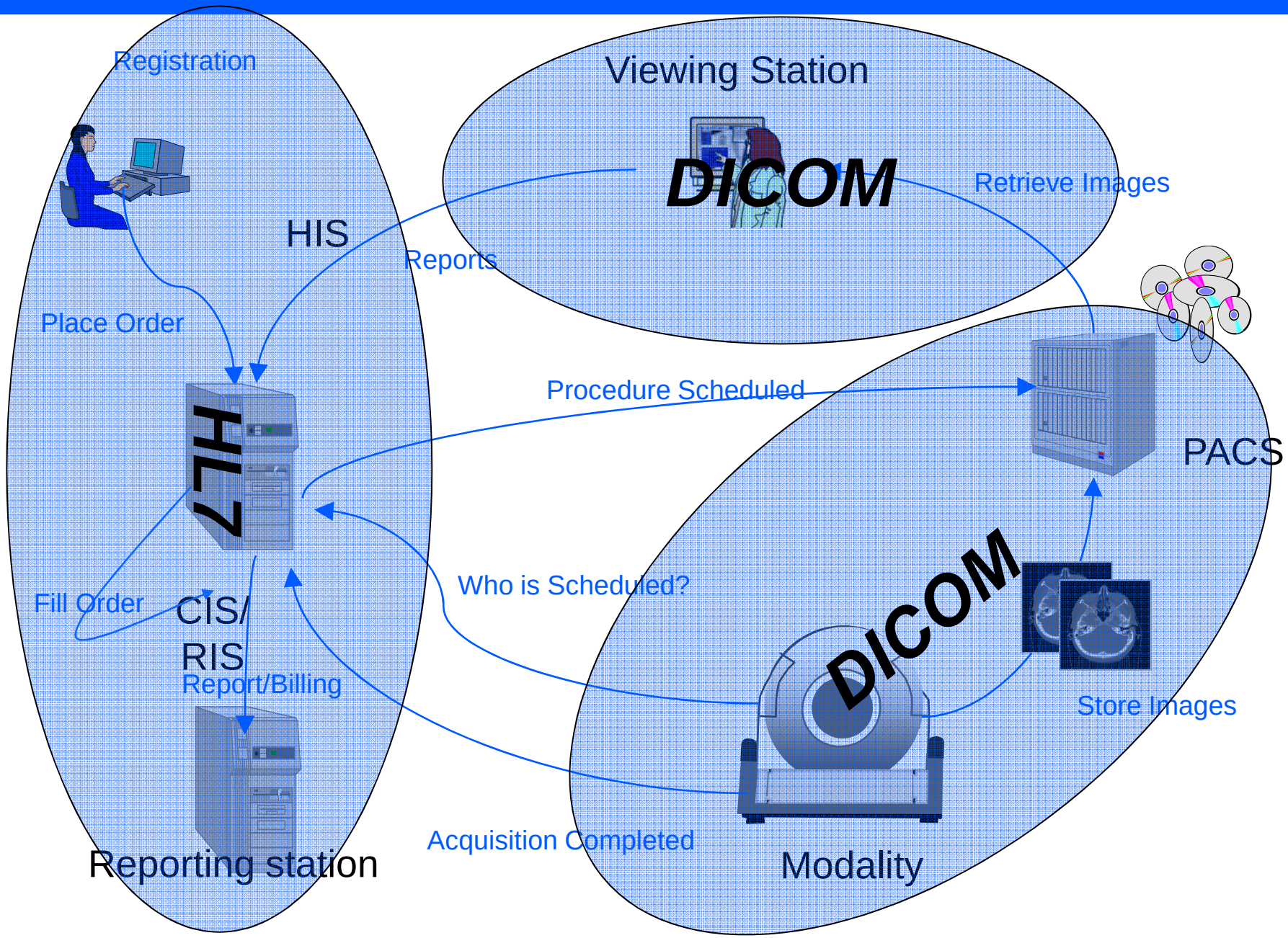


Standards for Healthcare

Health Level Seven®

DICOM℠

Digital Imaging and Communications in Medicine


Registration
HIS
Place Order
HL7
Fill Order
CIS/
RIS
Report/Billing
Reporting station
Viewing Station
DICOM
Reports
Retrieve Images
Procedure Scheduled
PACS
Who is Scheduled?
DICOM
Store Images
Acquisition Completed
Modality

History of DICOM

- SPI – Late 1980s
- ACR-NEMA 1.0 in 1985
- ACR-NEMA 2.0 in 1988
- DICOM 3.0 in 1992-1993

- Continuous development since then – DICOM 2011 is out

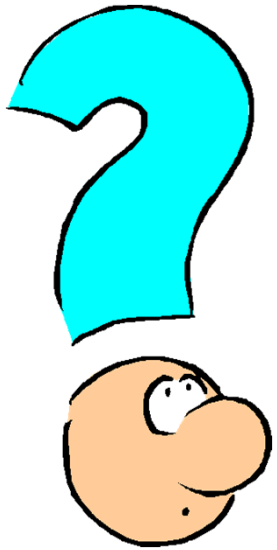
What is DICOM

- Standard for the communication and storage of medical images and associated data

- Defines
  - A Set of protocols to be followed by devices claiming conformance to the standard

  - Semantics & Syntax of commands and associated information which can be exchanged

  - Conformance Requirements for Implementations

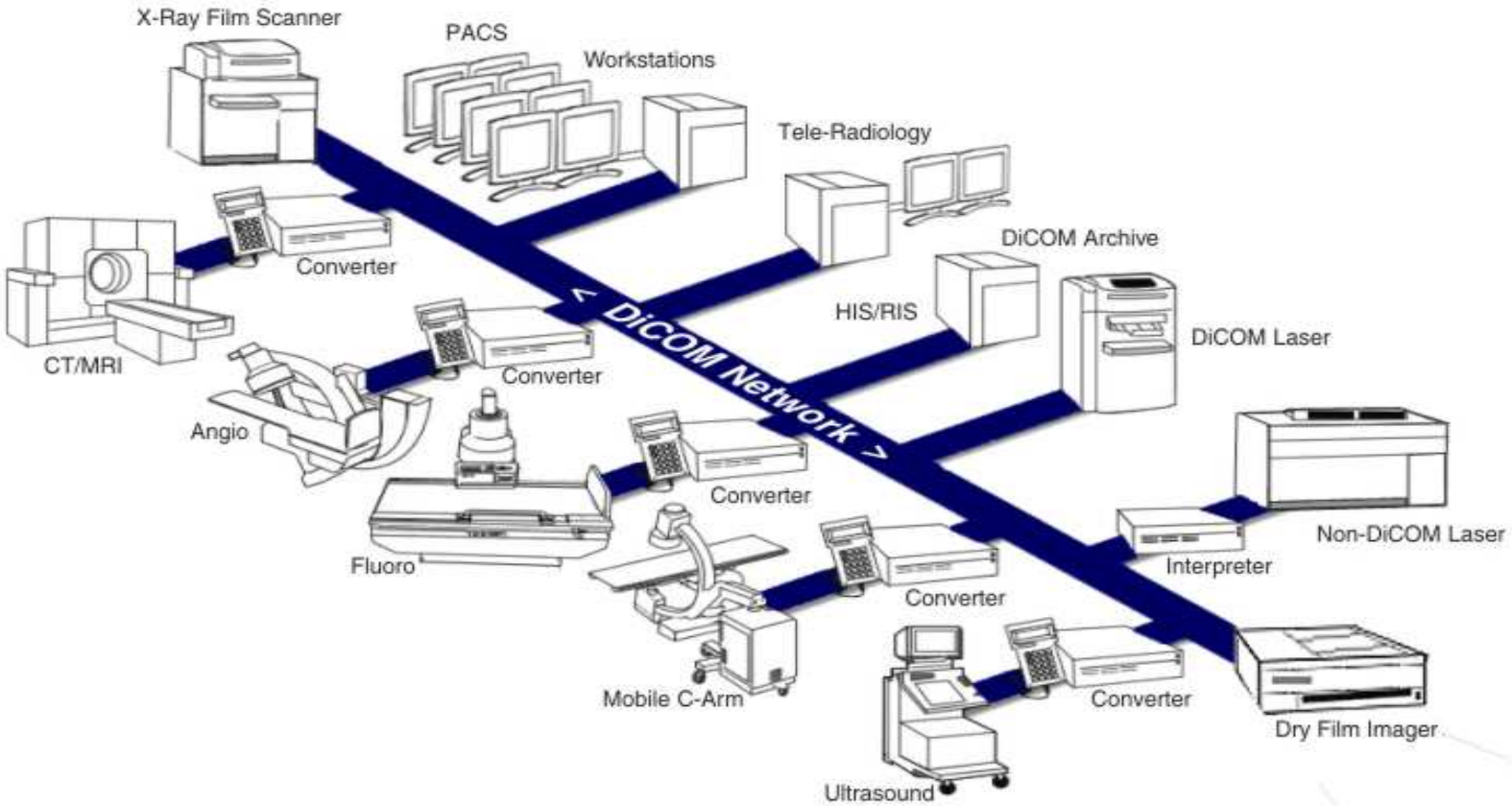


DICOM Is NOT

- Does not Define
  - Implementation details of any features
  - Overall set of features and functions to be expected
  - Testing and validation procedure to assess conformance

DICOM Conformance doesn't mean Interoperability

Areas of Application



Stakeholders

PHILIPS
sense and simplicity

NEMA
Setting Standards for Excellence

GE Healthcare

AGFA HealthCare

MIT MEDICAL IMAGING & TECHNOLOGY ALLIANCE
A DIVISION OF NEMA

JOHNS HOPKINS MEDICINE

SIEMENS medical

MERGE

AMERICAN COLLEGE OF CARDIOLOGY

EUROPEAN SOCIETY OF CARDIOLOGY®

SIIM

acr American College of Radiology

HIMSS The Source for Healthcare Information

SFR Société Française de Radiologie

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